HW III

EE 588: Optimization for the information and data sciences

University of Southern California

Assigned on: October 13, 2026 Due date: beginning of class on October 21, 2026

The following exercises from the text: 4.15, 4.17, 5.5 and 5.13 and the following additional exercises

- *Reformulating constraints in CVX.* Each of the following CVX code fragments describes a convex constraint on the scalar variables `x`, `y`, and `z`, but violates the CVX rule set, and so is invalid. Briefly explain why each fragment is invalid. Then, rewrite each one in an equivalent form that conforms to the CVX rule set. In your reformulations, you can use linear equality and inequality constraints, and inequalities constructed using CVX functions. You can also introduce additional variables, or use LMIs. Be sure to explain (briefly) why your reformulation is equivalent to the original constraint, if it is not obvious.

  Check your reformulations by creating a small problem that includes these constraints, and solving it using CVX. Your test problem doesn't have to be feasible; it's enough to verify that CVX processes your constraints without error.

  Remark. This *looks* like a problem about 'how to use CVX software', or 'tricks for using CVX'. But it really checks whether you understand the various composition rules, convex analysis, and constraint reformulation rules.

  1. `norm([x + 2*y, x - y]) == 0`
  2. `square(square(x + y)) <= x - y`
  3. `1/x + 1/y <= 1; x >= 0; y >= 0`
  4. `norm([max(x,1), max(y,2)]) <= 3*x + y`
  5. `x*y >= 1; x >= 0; y >= 0`
  6. `(x + y)^2/sqrt(y) <= x - y + 5`
  7. `x^3 + y^3 <= 1; x >= 0; y >= 0`
  8. `x + z <= 1 + sqrt(x*y - z^2); x >= 0; y >= 0`

- *Optimal activity levels.* Solve the optimal activity level problem described in exercise 4.17 in *Convex Optimization*, for the instance with problem data

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 3 & 1 & 1 \\ 2 & 1 & 2 & 5 \\ 1 & 0 & 3 & 2 \end{bmatrix}, \qquad c^{\max} = \begin{bmatrix} 100 \\ 100 \\ 100 \\ 100 \\ 100 \end{bmatrix}, \qquad p = \begin{bmatrix} 3 \\ 2 \\ 7 \\ 6 \end{bmatrix}, \qquad p^{\text{disc}} = \begin{bmatrix} 2 \\ 1 \\ 4 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 4 \\ 10 \\ 5 \\ 10 \end{bmatrix}.$$

  You can do this by forming the LP you found in your solution of exercise 4.17, or more directly, using CVX. Give the optimal activity levels, the revenue generated by each one, and

the total revenue generated by the optimal solution. Also, give the average price per unit for each activity level, i.e. the ratio of the revenue associated with an activity, to the activity level. (These numbers should be between the basic and discounted prices for each activity.) Give a *very brief* story explaining, or at least commenting on, the solution you find.